

Ryan Prototype Engineering Notes

Current Situation

I am unemployed, but I have time, tools, experience, and the ability to build.

The goal is not to find another engineering salary.

The goal is to package engineering capability into products and services that create significantly more value than hourly engineering work.

I enjoy building. The challenge is turning that activity into a repeatable business.

Key Realization

The gadgets are demos.

The tools are the products.

Examples:

- Custom pulley = demo
- Pulley generator = product
- SBC enclosure = demo
- Enclosure generator = product
- Gauge reader = demo
- Vision framework = product
- Sensor gateway = demo
- Industrial integration platform = product

The physical artifact is evidence.

The repeatable system behind the artifact is the business.

What I Actually Sell

Not:

- Embedded engineer
- Mechanical engineer
- Firmware engineer
- Consultant

Instead:

"I can turn ideas into working prototypes quickly."

People do not buy engineering.

People buy reduced uncertainty.

Customers want answers to:

- Can this be built?
- How much will it cost?
- What are the risks?
- Is the idea viable?

A prototype answers those questions.

Core Capability Stack

Mechanical

Capabilities:

- CAD
- OpenSCAD
- Fusion 360
- 3D printing
- Functional mechanisms
- Gears
- Pulleys
- Mounting systems
- Enclosures

Deliverable:

Functional mechanical hardware.

Electronics

Capabilities:

- Sensors
- Power systems
- Motor control
- Embedded hardware
- PCB design

Deliverable:

Working electronics.

Embedded

Capabilities:

- Embedded Linux
- BeagleBone
- Raspberry Pi
- IMX8M
- Firmware
- Drivers
- Communications

Deliverable:

Working systems.

Systems Integration

Capabilities:

- Hardware
- Software
- Mechanical
- Networking
- Vision
- Automation

Deliverable:

Working prototypes.

This is where the value is.

Competitive Advantage

Many people can do one of these:

- Mechanical
- Electronics
- Software

Fewer can do two.

Very few can do all three.

The real value is:

- Mechanical
- Electronics
- Embedded
- Integration

in a single person.

Productized Service Ideas

Feasibility Sprint

Fixed price.

Deliverables:

- Architecture
- Risk assessment
- Initial CAD
- Initial firmware
- Prototype plan

Timeline:

1–2 weeks

Purpose:

Answer whether a concept is viable.

Prototype Sprint

Fixed price.

Deliverables:

- Functional prototype
- CAD
- Firmware
- BOM
- Build instructions

Timeline:

2–6 weeks

Purpose:

Create a working demonstration.

Technical Rescue

High hourly rate.

Purpose:

Projects that are stuck.

Examples:

- Linux bring-up
 - Camera integration
 - Embedded debugging
 - Hardware troubleshooting
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Business Principles

Sell Outcomes

Bad:

40 hours of engineering.

Good:

Working prototype in two weeks.

Sell Certainty

Customers pay for answers.

Not effort.

Sell Speed

Speed is the differentiator.

Many engineers can do the work.

Fewer can do it quickly.

Demo Strategy

Every project should become evidence.

Structure:

Problem

What needed to be solved?

Constraints

What limitations existed?

Design Process

How was it solved?

Result

What worked?

Lessons Learned

What would be done differently?

Example: Encoder Pulley

Problem:

Need to connect shafts 72 mm apart.

Constraints:

- S3M belt
- D shaft
- 3D printable
- Encoder application

Engineering decisions:

- Belt instead of gears
- 41T pulley
- D bore
- Chamfered flanges
- Lightening holes
- Bearing offset

Result:

Functional pulley.

The pulley is not the story.

The decision-making process is the story.

Potential Demonstration Projects

Analog Gauge Reader

Demonstrates:

- Vision
- AI
- Embedded Linux
- Electronics
- Mechanical design

Strong because people immediately understand it.

Sensor Gateway

Demonstrates:

- Embedded
 - Networking
 - Industrial protocols
 - Dashboard development
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Camera System

Demonstrates:

- Embedded Linux
 - Hardware integration
 - Vision systems
 - Edge AI
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Parametric Mechanical Components

Demonstrates:

- CAD
- Automation
- Manufacturing

Potential product.

Tool Opportunities

Pulley Generator

Input:

- Belt type
- Shaft size
- Center distance
- Bore type

Output:

- STL
 - STEP
 - CAD
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Enclosure Generator

Input:

- SBC
- Connector locations
- Mounting requirements

Output:

Manufacturable enclosure.

Mechanism Generator

Input:

- Motion requirements
- Shaft spacing
- Reduction ratio

Output:

Complete assembly.

Marketing Notes

The internet rewards visible builders.

Publish:

- GitHub
- Hackaday
- Reddit
- LinkedIn
- YouTube

Not because any single post matters.

Because over time the body of work becomes proof.

Important Insight

The prototype is not the product.

The prototype is evidence.

The customer buys confidence.

The customer buys reduced risk.

The customer buys speed.

Long-Term Vision

Create a system where engineering intent becomes hardware.

Input:

"I need two shafts connected."

Output:

- CAD
- BOM
- STL

- Drawings
- Prototype

The end goal is not becoming a better CAD designer.

The end goal is reducing the time between idea and working hardware.